

Theoretical Neuroscience : Chapter 1

Gyuyoung Hwang

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Seoul National University
hgy0407@snu.ac.kr

Objective : Basic statistics describing how stimulus evoke spike sequences.

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Structure of a Neuron

Spike trains and Firing rates

What makes a neuron fire?

Stimulus that triggers spikes

Neural Coding

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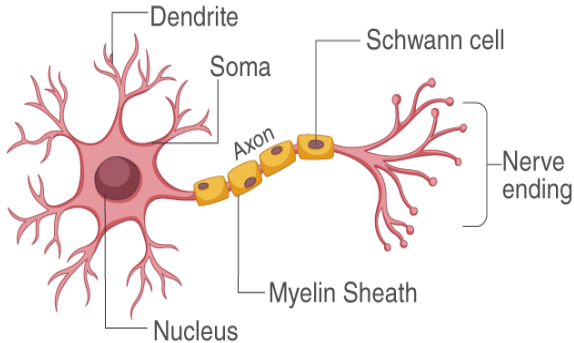
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Structure of a Neuron

STRUCTURE OF NEURON

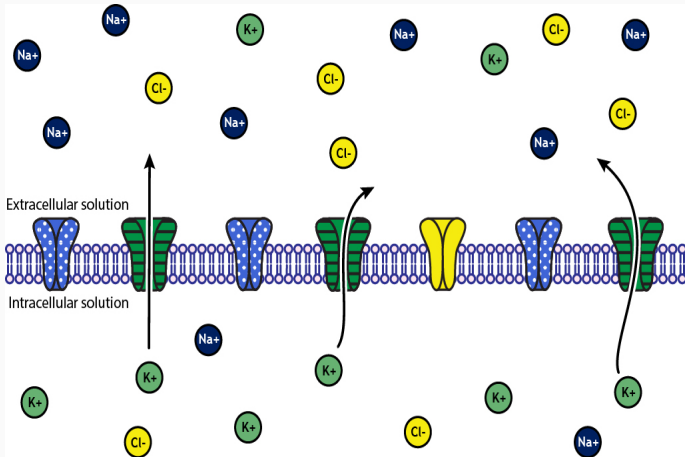


From <https://byjus.com/biology/neurons/>

Axons and dendrites

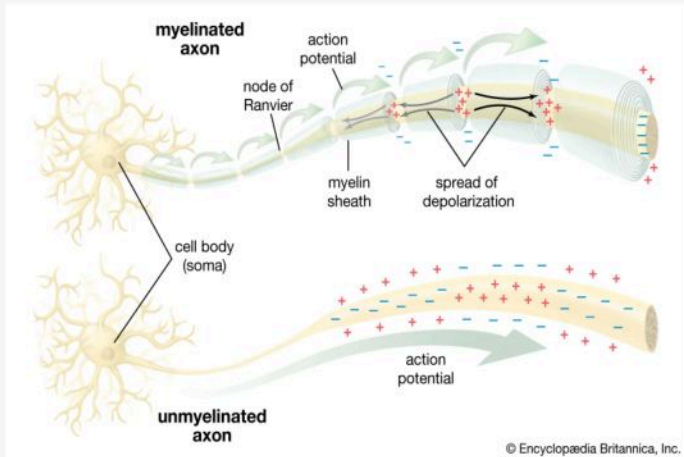
- One axon per one neuron **vs** many dendrites per one neuron.
- Axon is long **vs** dendrites are short.
- Dendrites receive messages from other cells.
- Axons pass messages from the cell body to other neurons or muscles.
- Myelin Sheath covers the axon and it helps speed the neural impulses.

Action Potential



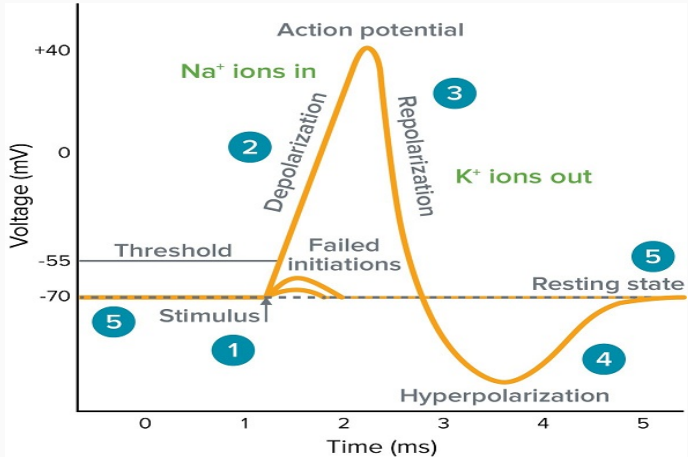
From <https://www.britannica.com/science/action-potential>

Action Potential



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Action Potential



From <https://www.moleculardevices.com/applications/patch-clamp-electrophysiology/what-action-potential>

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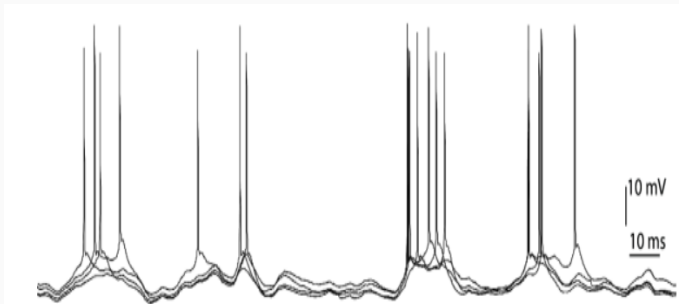
Because of the action potential, there arise a sequence of spikes represented by the dirac delta functions, which is called a **neural response function**

$$\rho(t) = \sum_{i=1}^n \delta(t - t_i).$$

For any well-behaved function h , we will use the identity

$$\sum_{i=1}^n h(t - t_i) = \int_{-\infty}^{\infty} d\tau h(\tau) \rho(t - \tau).$$

Spikes



From <https://neurondynamics.epfl.ch/online/Ch7.S1.html>

Repetition by four stimuli

We introduce several terminologies. Keep in mind that

Stimulus $\rightarrow$ Firing $\rightarrow$ Spike

- Spike-count rate r .

$$r = \frac{n}{T} = \frac{1}{T} \int_0^T d\tau \rho(\tau).$$

Thus, the **the spike-count rate** is the time average of **the neural response function** over the duration of the trial.

- Averaging over the trials that use the same stimulus

$$\langle \rho(t) \rangle.$$

- Average firing rate $\langle r \rangle$

$$\langle r \rangle = \frac{\langle n \rangle}{T} = \frac{1}{T} \int_0^T d\tau \langle \rho(\tau) \rangle = \frac{1}{T} \int_0^T dt r(t).$$

- Firing rate $r(t)$

$$r(t) = \frac{1}{\Delta t} \int_t^{t+\Delta t} d\tau \langle \rho(\tau) \rangle.$$

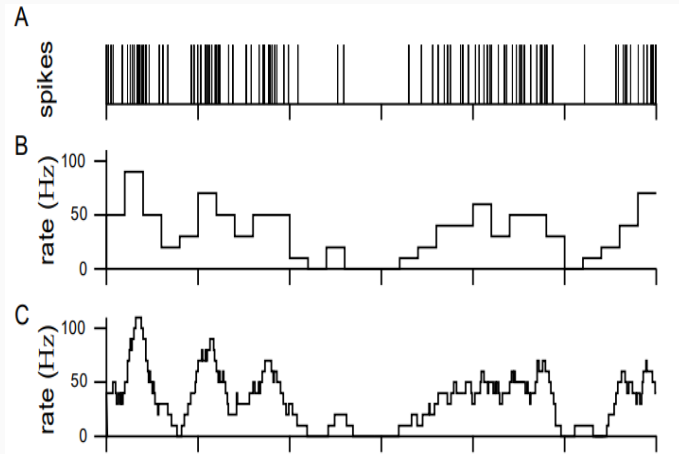
Observe that in the limit $\Delta t \rightarrow 0$, we have $r(t) = \langle \rho(t) \rangle$. However, Δt should be sufficiently large so that there are sufficient numbers of spikes within the interval. However, we will use that for a good enough function h , we have

$$\int d\tau h(\tau) \langle \rho(t - \tau) \rangle = \int d\tau h(\tau) r(t - \tau).$$

- $r(t)\Delta t$ is the probability that a spike occurs during the time interval whose length is Δt .

Measuring Firing Rates

- How to measure the firing rate $r(t)$?



Measuring Firing Rates

- B. Divide the time interval with equal size, and count.
- Or, we approximate $r(t)$ by using the kernel as follows:

$$r_a(t) = \int_{-\infty}^{\infty} d\tau \omega(\tau) \rho(t - \tau),$$

which is called a linear filter with the filter kernel ω .

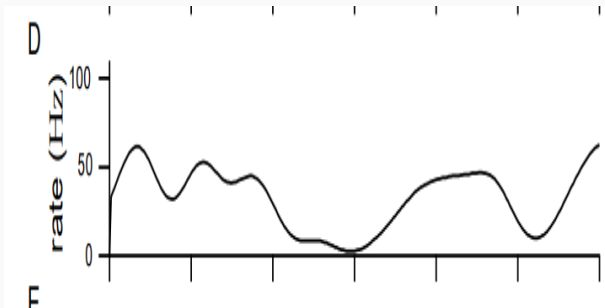
- C. Take the window function as

$$\omega(t) = \begin{cases} 1/\Delta t & \text{if } -\Delta t/2 \leq t < \Delta t/2 \\ 0 & \text{otherwise.} \end{cases}$$

- Or, we may take

$$\omega(\tau) = \frac{1}{\sqrt{2\pi}\sigma_\omega} \exp\left(-\frac{\tau^2}{2\sigma_\omega^2}\right),$$

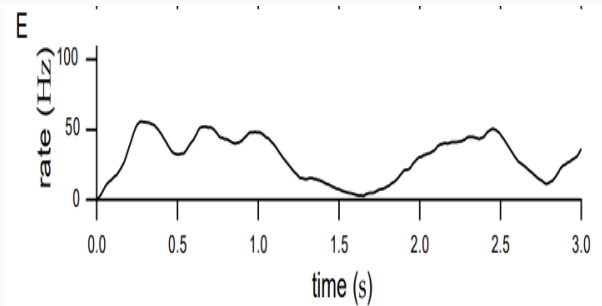
which gives the approximation



- Or, if we only take the past effect, we take

$$\omega(\tau) = [\alpha^2 \tau \exp(-\alpha \tau)]_+,$$

which gives the approximation



Tuning Curves

- The neural response tuning curve:

The average firing rate written as a function of s , $\langle r \rangle = f(s)$.

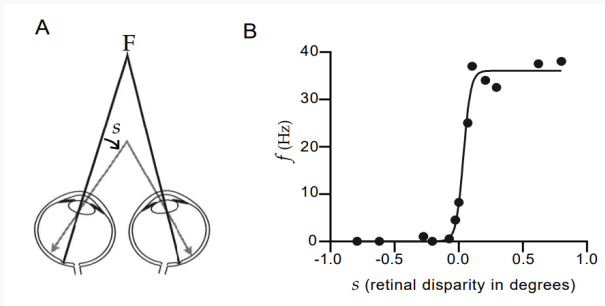
- For example,

$$f(s) = r_{max} \exp\left(-\frac{1}{2} \left(\frac{s - s_{max}}{\sigma_f}\right)^2\right),$$

- We assume that the stimulus function s is periodic in period T .

Tuning Curves

- Consider the retinal disparity,



Here, we take

$$f(s) = \frac{r_{\max}}{1 + \exp((s_{1/2} - s)/\Delta_s)}.$$

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- **Spike-triggered average stimulus**, $C(\tau)$, is the average value of the stimulus a time interval τ before a spike is fired.

$$C(\tau) = \left\langle \frac{1}{n} \sum_{i=1}^n s(t_i - \tau) \right\rangle \approx \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^n s(t_i - \tau) \right\rangle,$$

where we assumed $n \approx \langle n \rangle$. Observe that

$$\begin{aligned} C(\tau) &= \frac{1}{\langle n \rangle} \int_0^T dt \langle \rho(t) \rangle s(t - \tau) \\ &= \frac{1}{\langle n \rangle} \int_0^T dt r(t) s(t - \tau). \end{aligned}$$

- The **correlation function** of the firing rate and the stimulus is

$$Q_{rs}(\tau) = \frac{1}{T} \int_0^T dt \, r(t)s(t + \tau).$$

Now, using that $\langle r \rangle = \langle n \rangle / T$, we see that

$$C(\tau) = \frac{1}{\langle r \rangle} Q_{rs}(-\tau),$$

therefore, the spike-triggered average stimulus $C(\tau)$ is essentially the reverse correlation function.

White-noise stimuli

- The **white-noise stimulus** has the defining characteristic that its value at any one time is uncorrelated with its value at any other time. Define the **stimulus autocorrelation function**

$$Q_{ss}(\tau) = \frac{1}{T} \int_0^T dt s(t)s(t+\tau).$$

For the white noise, the autocorrelation function is 0 in the range $-T/2 < \tau < T/2$ except when $\tau = 0$, i.e.

$$Q_{ss}(\tau) = \sigma_s^2 \delta(\tau).$$

This is equivalent to saying that white noise has equal power at all frequencies.

The Power spectrum of White noise

The Fourier transform of the stimulus autocorrelation function Q_{ss} is

$$\hat{Q}_{ss}(\omega) = \frac{1}{T} \int_{-T/2}^{T/2} d\tau Q_{ss}(\tau) \exp(i\omega\tau),$$

which is called the power spectrum. For the white noise $Q_{ss}(\tau) = \sigma_s^2 \delta(\tau)$,

$$\hat{Q}_{ss}(\omega) = \frac{\sigma_s^2}{T} \int_{-T/2}^{T/2} d\tau \delta(\tau) \exp(i\omega\tau) = \frac{\sigma_s^2}{T}.$$

The Power spectrum of White noise

In general, consider the Fourier transform of the stimulus

$$\hat{s}(\omega) = \frac{1}{T} \int_0^T d\tau s(t) \exp(i\omega t).$$

Then, we compute that

$$\begin{aligned}\hat{Q}_{ss}(\omega) &= \frac{1}{T} \int_0^T dt s(t) \frac{1}{T} \int_{-T/2}^{T/2} d\tau s(t + \tau) \exp(i\omega\tau) \\ &= \frac{1}{T} \int_0^T dt s(t) \exp(-i\omega t) \frac{1}{T} \int_{-T/2}^{T/2} d\tau s(t + \tau) \exp(i\omega(t + \tau)) \\ &= \bar{\hat{s}}(\omega) \hat{s}(\omega) = |\hat{s}(\omega)|^2.\end{aligned}$$

In practice, we instead estimate $\hat{Q}_{ss}(\omega)$ by

$$\frac{\Delta T}{T} \left| \sum_{m=1}^{T/\Delta t} s(t) \exp(-i\omega m \Delta t) \right|^2.$$

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- We want to know the probabilities corresponding to every sequence of **spikes** that can be evoked by the **stimulus**.
- The probability that a sequence of n spikes occurs with spike i falling between times t_i and $t_i + \Delta t$ for each $i = 1, 2, \dots, n$ is given by the density as

$$P[t_1, \dots, t_n] = p[t_1, \dots, t_n](\Delta t)^n.$$

- We consider the Poisson process and separate two cases depending on whether the firing rate $r(t)$ is constant over time or not. (Homogeneous/Inhomogeneous).

Homogeneous Poisson process

- In this case $r(t) = r$.
- Then, the probability $P[t_1, \dots, t_n]$ is independent of the explicit numbers t_i . Thus, if

$$0 \leq t_1 \leq t_2 \leq \dots \leq t_n \leq T,$$

then, we have the relationship

$$P[t_1, \dots, t_n] = n! P_T[n] \left(\frac{\Delta t}{T} \right)^n,$$

where $P_T[n]$ is the probability that an arbitrary sequence of exactly n spikes occurs within a trial of duration T .

- Indeed, we can deduce that

$$P_T[n] = \frac{(rT)^n}{n!} \exp(-rT).$$

Derivation of $P_T[n]$

1. Divide the time T into M bins of size $\Delta t = T/M$. Assume that Δt is small enough so that we never get two spikes within any one bin.
2. Note that $P_T[n]$ is the product of : the probability of generating n spikes within a specified set of the M bins, the probability of not generating spikes in the remaining $M - n$ bins, and a combinatorial factor equal to the number of ways of putting n spikes into M bins.
3. Thus,

$$P_T[n] = \lim_{\Delta t \rightarrow 0} \binom{M}{n} (r\Delta t)^n (1 - r\Delta t)^{M-n}$$

4. In $\Delta t \rightarrow 0$ with $M\Delta t = T$, we approximate $M - n \approx M = T/\Delta t$.
Then,

$$\lim_{\Delta t \rightarrow 0} (1 - r\Delta t)^{M-n} = \exp(-rT).$$

Homogeneous Poisson process

- Using $P_T[n]$, we compute **the variance of the spike count** for spikes counted over an interval of duration T :

$$\sigma_n^2 = \langle n^2 \rangle - \langle n \rangle^2 = rT.$$

Sketch of proof) We note that

$$\langle n \rangle = \sum_{n=0}^{\infty} n P_T[n] = \sum_{n=0}^{\infty} \frac{n(rT)^n}{n!} \exp(-rT),$$

and also,

$$\sigma_n^2(T) = \sum_{n=0}^{\infty} n^2 P_T[n] - \langle n \rangle^2.$$

Computing moments

Using the moment generating function

$$g(\alpha) := \sum_{n=0}^{\infty} \frac{(rT)^n \exp(\alpha n)}{n!} \exp(-rT).$$

We observe that

$$g(\alpha) = \exp(-rT) \exp(rT e^{\alpha}).$$

Thus,

$$\begin{aligned} \frac{dg}{d\alpha} &= rT e^{\alpha} \exp(rT e^{\alpha}) \exp(-rT), \\ \frac{d^2g}{d\alpha^2} &= (rT)^2 e^{2\alpha} \exp(rT e^{\alpha}) \exp(-rT) + rT e^{\alpha} \exp(rT e^{\alpha}) \exp(-rT). \end{aligned}$$

Inhomogeneous Poisson Statistics

- When $r(t)$ depends on time, different sequences of n spikes occur with different probabilities and $p[t_1, \dots, t_n]$ depends on the spike times.

$$p[t_1, \dots, t_n] = \exp\left(-\int_0^T dt r(t)\right) \prod_{i=1}^n r(t_i).$$

Sketch of derivation) First, we compute the probability that **no spikes are generated** during the time interval from t_i to t_{i+1} . Divide the time interval by $M\Delta t = t_{i+1} - t_i$. Then observe that

$$P[\text{no spikes}] = \prod_{m=1}^M (1 - r(t_i + m\Delta t)\Delta t).$$

Take logarithm and use $\ln(1 - x) \approx -x$. Then as $\Delta t \rightarrow 0$.

$$\ln P[\text{no spikes}] = -\sum_{m=1}^M r(t_i + m\Delta t)\Delta t \rightarrow -\int_{t_i}^{t_{i+1}} dt r(t).$$

Finally, we have the probability density

$$\begin{aligned} p[t_1, \dots, t_n] &= \exp\left(-\int_0^{t_1} dtr(t)\right) \exp\left(-\int_{t_n}^T dtr(t)\right) r(t_n) \\ &\quad \times \prod_{i=1}^{n-1} r(t_i) \exp\left(-\int_{t_i}^{t_{i+1}} dtr(t)\right) \\ &= \exp\left(-\int_0^T dtr(t)\right) \prod_{i=1}^n r(t_i). \end{aligned}$$

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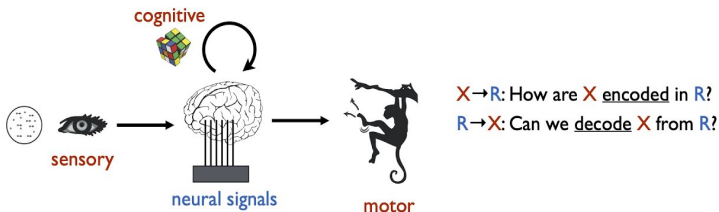
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Neural coding problem

How is information represented in the *short* time scale?



From <https://www.youtube.com/watch?v=DIFxUEdGImQ>

- The nature of the neural code is a topic of intense debate within the neuroscience community.
- Whether neurons use **rate coding** vs **tempopral coding**.
- That is, individual action potentials and neurons encode independently of each other **vs** correlations between different spikes and neurons carry significant amounts of information.
- However, both happens in reality.

- Independent-spike code vs Correlation code
- The neural response and its relation to the stimulus are completely characterized by the probability distribution of spike times as a function of the stimulus.
- If spike generation is described by an inhomogeneous Poisson process, then

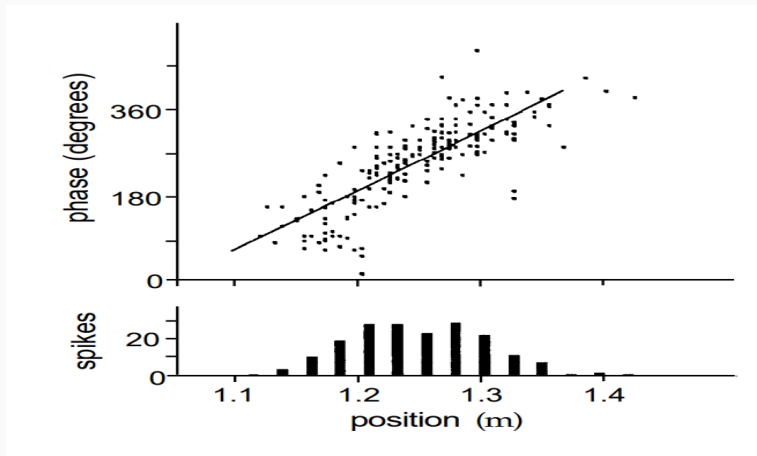
$$p[t_1, \dots, t_n] = \exp\left(-\int_0^T dt r(t)\right) \prod_{i=1}^n r(t_i),$$

where $r(t)$ is the time-dependent firing rate.

Independent-spike code vs Correlation code

- In this case, $r(t)$ contains all the information about the stimulus that can be extracted from the spike train.
⇒ Independent-spike code.
- If individual spikes do not encode independently of each other, we call the neural code
⇒ Correlation code.
- Independent-spike coding is regarded as quite convincing at least as a fairly accurate approximation.
- However, this does not mean that we do not consider the **ensemble** of neurons.

Neural Coding



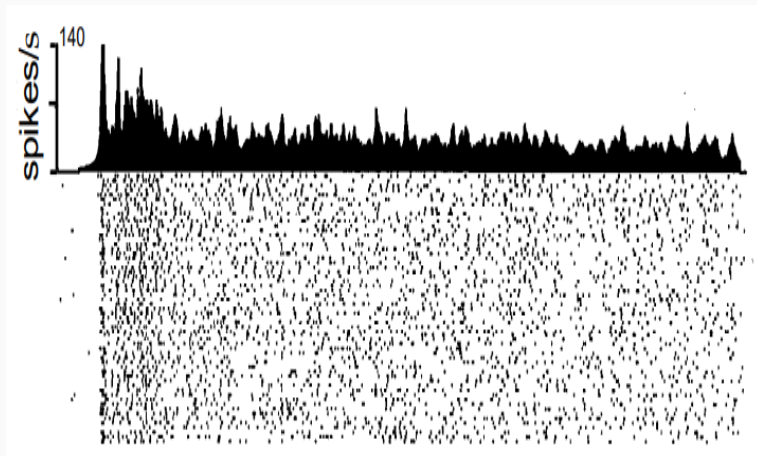
Synchronous firing in the rat hippocampus

However, synchronization does not mean spikes are correlated.

Temporal Codes

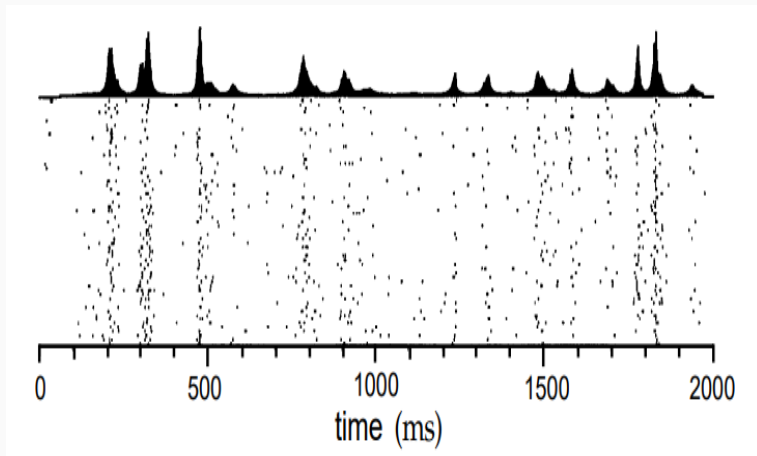
- When **precise spike timing** or **high-frequency firing-rate fluctuations** are found to carry information, the neural code is often identified as a **temporal code**.
- The temporal coding is distinct and independent from the issue of independent-spike coding.
- We distinguish temporal coding from rate coding according to whether the firing rate $r(t)$ changes slowly or rapidly.

rate coding vs temporal coding



When $r(t)$ varies slowly, rate code.

rate coding vs temporal coding



When $r(t)$ varies rapidly, temporal code.

- Q. how to distinguish rate from temporal?
 1. Use spikes to distinguish slow from rapid.

Each peak corresponds to the firing of only one action potential.
Spike frequency = Peak frequency.
 2. Use stimulus.

Spike timing scale $<$ fastest time for variations of the stimulus.
- The central challenge is to **identify relationships** between the firing patterns of different neurons in a responding population and to understand their significance for neural coding.

Thank you for the listening!